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Embedded Systems and IOT - Lab Exercise: Schematics with ESP32

Part A:

1. Total resistance in ohms:
 - a. **100 ohms**
 - b. **12.5 Ohms**
2. Total current from power supply:
 - a. **1.25A**
 - b. **10A**
3. Current through each resistor:
 - a. **1.25A for each resistor**
 - b. 20 Ohm – **6.25A**
100 Ohm – **1.25A**
50 Ohm – **2.5A**
4. Voltage drop through each resistor:
 - a. 20 Ohm – **25V**
30 Ohm – **37.5V**
50 Ohm – **62.5V**
 - b. **125V for each resistor**
5. Power dissipated through each resistor:
 - a. 20 Ohm – **31.25W**
30 Ohm – **46.875W**
50 Ohm – **78.125W**
 - b. 20 Ohm – **781.25W**
100 Ohm – **156.25W**
50 Ohm – **312.5W**